

Emerging roles of microglia cells in the regulation of adult neural stem cells.

PMID: 28702478 | Indexed in PubMed

Microglia cells were first described as a component for the brain with few beneficial functions. The classical point of view implied that these cells had inflammatory properties more than benefits for brain homeostasis. To date, this assumption has changed and new roles of microglia cells are continuously discovered. Although, the main function of microglia cells is to provide a cellular defense against harmful or pathogen agents (bacteria, viruses, fungi, toxins, etc.), recent evidence indicates that microglial cells are dynamic modulators of synaptic pruning, brain development and neurogenesis by maintaining a balance of local cell population. In this commentary, we summarized the emerging role of the relationship between microglia cells and the neural stem cells resident in the ventricular-subventricular zone (V-SVZ), the largest neurogenic niche in the adult brain.

Neuroglia Cells Transcriptomic in Brain Development, Aging and Neurodegenerative Diseases.

PMID: 36818562 | Indexed in PubMed

Glia cells are essential for brain functioning during development, aging and disease. However, the role of astroglia plays during brain development is quite different from the role played in the adult lesioned brain. Therefore, a deeper understanding of pathomechanisms underlying astroglia activity in the aging brain and cerebrovascular diseases is essential to guide the development of new therapeutic strategies. To this end, this review provides a comparison between the transcriptomic activity of astroglia cells during development, aging and neurodegenerative diseases, including cerebral ischemia. During fetal brain development, astrocytes and microglia often affect the same developmental processes such as neuro-/gliogenesis, angiogenesis, axonal outgrowth, synaptogenesis, and synaptic pruning. In the adult brain astrocytes are a critical player in the synapse remodeling by mediating synapse elimination while microglia activity has been associated with changes in synaptic plasticity and remove cell debris by constantly sensing the environment. However, in the lesioned brain astrocytes proliferate and play essential functions with regard to energy supply to the neurons, neurotransmission and buildup of a protective scar isolating the lesion site from the surroundings. Inflammation, neurodegeneration, or loss of brain homeostasis induce changes in microglia gene expression, morphology, and function, generally referred to as "primed" microglia. These changes in gene expression are characterized by an enrichment of phagosome, lysosome, and antigen presentation signaling pathways and is associated with an up-regulation of genes encoding cell surface receptors. In addition, primed microglia are characterized by upregulation of a network of genes in response to interferon gamma. Conclusion. A comparison of astroglia cells transcriptomic activity during brain development, aging and neurodegenerative disorders might provide us with new therapeutic strategies with which to protect the aging brain and improve clinical outcome.

Microglia-neuron interactions in schizophrenia.

PMID: 38510107 | Indexed in PubMed

Multiple lines of evidence implicate increased neuroinflammation mediated by glial cells to play a key

role in neurodevelopmental disorders such as schizophrenia. Microglia, which are the primary innate immune cells of the brain, are crucial for the refinement of the synaptic circuitry during early brain development by synaptic pruning and the regulation of synaptic plasticity during adulthood. Schizophrenia risk factors as genetics or environmental influences may further be linked to increased activation of microglia, an increase of pro-inflammatory cytokine levels and activation of the inflammasome resulting in an overall elevated neuroinflammatory state in patients. Synaptic loss, one of the central pathological hallmarks of schizophrenia, is believed to be due to excess removal of synapses by activated microglia, primarily affecting glutamatergic neurons. Therefore, it is crucial to investigate microglia-neuron interactions, which has been done by multiple studies focusing on post-mortem brain tissues, brain imaging, animal models and patient iPSC-derived 2D culture systems. In this review, we summarize the major findings in patients and in vivo and in vitro models in the context of neuron-microglia interactions in schizophrenia and secondly discuss the potential of anti-inflammatory treatments for the alleviation of positive, negative, and cognitive symptoms.

In Vivo Two-Photon Imaging of Microglial Synapse Contacts.

PMID: 31392692 | Indexed in PubMed

Microglia are traditionally known as immune sentinels of the brain and as key player in the pathogenesis of neurodegenerative diseases such as Alzheimer's disease, Parkinson disease, or amyotrophic lateral sclerosis. Recently, they were also identified as synaptic organizer, promoting formation and maturation of synapses as well as modifying synaptic activity. Interestingly, microglia-mediated synaptic pruning and microglia-mediated changes in synaptic plasticity were observed both during brain development and in neurodegenerative diseases, stressing the key role of microglia-synapse interaction in these processes. Here we described a technique for noninvasive in vivo monitoring of microglia-synapse interactions by means of two-photon microscopy.

Strategies and Tools for Studying Microglial-Mediated Synapse Elimination and Refinement.

PMID: 33708226 | Indexed in PubMed

The role of microglia in controlling synapse homeostasis is becoming increasingly recognized by the scientific community. In particular, the microglia-mediated elimination of supernumerary synapses during development lays the basis for the correct formation of neuronal circuits in adulthood, while the possible reactivation of this process in pathological conditions, such as schizophrenia or Alzheimer's Disease, provides a promising target for future therapeutic strategies. The methodological approaches to investigate microglial synaptic engulfment include different in vitro and in vivo settings. Basic in vitro assays, employing isolated microglia and microbeads, apoptotic membranes, liposomes or synaptosomes allow the quantification of the microglia phagocytic abilities, while co-cultures of microglia and neurons, deriving from either WT or genetically modified mice models, provide a relatively manageable setting to investigate the involvement of specific molecular pathways. Further detailed analysis in mice brain is then mandatory to validate the in vitro assays as representative for the in vivo situation. The present review aims to dissect the main technical approaches to investigate microglia-mediated phagocytosis of neuronal and synaptic substrates in critical developmental time windows.

Work hard, play hard: how sexually differentiated microglia work to shape social play and reproductive behavior.

PMID: 36172465 | Indexed in PubMed

Microglia are brain-resident immune cells that play a critical role in synaptic pruning and circuit fine-tuning during development. In the adult brain, microglia actively survey their local environment and mobilize inflammatory responses to signs of damage or infection. Sex differences in microglial gene expression and function across the lifespan have been identified, which play a key role in shaping brain function and behavior. The levels of sex hormones such as androgens, estrogens, and progesterone vary in an age-dependent and sex-dependent manner. Microglia respond both directly and indirectly to changes in hormone levels, altering transcriptional gene expression, morphology, and function. Of particular interest is the microglial function in brain regions that are highly sexually differentiated in development such as the amygdala as well as the pre-optic and ventromedial hypothalamic regions. With a focus on hormone-sensitive developmental windows, this review compares male and female microglia in the embryonic, developing, and adult brain with a particular interest in the influence of sex hormones on microglial wiring of social, reproductive, and disordered behavior circuits in the brain.

Detection of Synaptic Proteins in Microglia by Flow Cytometry.

PMID: 33132837 | Indexed in PubMed

A growing body of evidence indicates that microglia actively remove synapses in vivo, thereby playing a key role in synaptic refinement and modulation of brain connectivity. This phenomenon was mainly investigated in immunofluorescence staining and confocal microscopy. However, a quantification of synaptic material in microglia using these techniques is extremely time-consuming and labor-intensive. To address this issue, we aimed to quantify synaptic proteins in microglia using flow cytometry. With this approach, we first showed that microglia from the healthy adult mouse brain contain a detectable level of VGLUT1 protein. Next, we found more than two-fold increased VGLUT1 immunoreactivity in microglia from the developing brain (P15) as compared to adult microglia. These data indicate that microglia-mediated synaptic pruning mostly occurs during the brain developmental period. We then quantified the VGLUT1 staining in microglia in two transgenic models characterized by pathological microglia-mediated synaptic pruning. In the 5xFAD mouse model of Alzheimer's disease (AD) microglia exhibited a significant increase in VGLUT1 immunoreactivity before the onset of amyloid pathology. Moreover, conditional deletion of TDP-43 in microglia, which causes a hyper-phagocytic phenotype associated with synaptic loss, also resulted in increased VGLUT1 immunoreactivity within microglia. This work provides a quantitative assessment of synaptic proteins in microglia, under homeostasis, and in mouse models of disease.

Microglia Increase Inflammatory Responses in iPSC-Derived Human BrainSpheres.

PMID: 30619100 | Indexed in PubMed

Human induced pluripotent stem cells (iPSCs), together with 21st century cell culture methods, have the

potential to better model human physiology with applications in toxicology, disease modeling, and the study of host-pathogen interactions. Several models of the human brain have been developed recently, demonstrating cell-cell interactions of multiple cell types with physiologically relevant 3D structures. Most current models, however, lack the ability to represent the inflammatory response in the brain because they do not include a microglial cell population. Microglia, the resident immunocompetent phagocytes in the central nervous system (CNS), are not only important in the inflammatory response and pathogenesis; they also function in normal brain development, strengthen neuronal connections through synaptic pruning, and are involved in oligodendrocyte and neuronal survival. Here, we have successfully introduced a population of human microglia into 3D human iPSC-derived brain spheres (BrainSpheres, BS) through co-culturing cells of the Immortalized Human Microglia - SV40 cell line with the BS model (μ BS). We detected an inflammatory response to lipopolysaccharides (LPS) and flavivirus infection, which was only elicited in the model when microglial cells were present. A concentration of 20 ng/mL of LPS increased gene expression of the inflammatory cytokines interleukin-6 (IL-6), IL-10, and IL-1 β , with maximum expression at 6-12 h post-exposure. Increased expression of the IL-6, IL-1 β , tumor necrosis factor alpha (TNF- α), and chemokine (C-C motif) ligand 2 (CCL2) genes was observed in μ BS following infection with Zika and Dengue Virus, suggesting a stronger inflammatory response in the model when microglia were present than when only astrocyte, oligodendrocyte, and neuronal populations were represented. Microglia innately develop within cerebral organoids (Nature communications), our findings suggest that the μ BS model is more physiologically relevant and has potential applications in infectious disease and host-pathogen interactions research.

Gestational Toxicity of Calabash Chalk (Nzu) in Wistar Rats.

PMID: 30598913 | Indexed in PubMed

Calabash chalk (Nzu) is a geophagic material, consumed by many pregnant women and breastfeeding mothers as remedy for morning sickness in most African countries. The present study aims to evaluate the impact of Calabash chalk consumption in pregnant Wistar rats. Median lethal dose of Calabash chalk was initially determined in Wistar rats to be >5000 mg/kg po using Lorke's method. Thereafter, 24 pregnant Wistar rats were divided randomly into 3 groups (n = 8) and orally administered 0, 400, or 800 mg/kg of Calabash chalk from gestation day 0 to 20. Maternal body weights were monitored during sample administration. Fetuses were delivered under anesthesia by cesarean section and pregnancy outcome was assessed. Calabash chalk exposure inhibited maternal weight gain. Uterine implantations were absent in 85% of Calabash chalk exposed rats, and the number of implantation sites were decreased (P = 0.0262) in the rest, compared to control. Uterine weight and the number of fetuses formed in uterine horns of Calabash chalk exposed pregnant rats were decreased (P = 0.0204) when compared with control. In addition, there was resorption of pregnancy, abortion (58%), and stillbirth (5%) in Calabash chalk exposed rats, and fetuses delivered were sluggish and pale. Most of these effects observed were dose-dependent. The results suggest that Calabash chalk has a negative relationship with maternal health and pregnancy outcome.

The effect of calabash chalk on some hematological parameters in female adult Wistar rats.

PMID: 27263602 | Indexed in PubMed

Calabash chalk is a naturally occurring mineral consumed among the Nigerian community for pleasure and commonly by pregnant women as a remedy for morning sickness. Reports have shown that it contains different toxic substances, with lead being the most abundant. This study was therefore undertaken to ascertain the effect of two commonly available preparations of this chalk on some hematological parameters. Twenty-four adult female Wistar rats with average weight of 100 g were assigned into three groups (1, 2, 3). Group 1 served as the control and the animals received distilled water, while Groups 2 and 3 were treated by oral gavage with 40 mg/kg of non-salted (NSCC) and salted calabash chalk (SCC), respectively, for 14 days. The hemoglobin (Hb) concentration and red blood cell (RBC) count were significantly ($p < 0.05$, 0.001 respectively) lower in the NSCC group, while erythrocyte sedimentation rate (ESR) was significantly ($p < 0.05$) higher in the NSCC group compared to the control. There were no significant differences in packed cell volume (PCV), white blood cell (WBC) and platelet (PI) counts compared to the control. The SCC group presented no significant difference in all blood count parameters compared to the control. This infers that calabash chalk, particularly the non-salted form, alters the normal concentration of Hb, RBC and PI counts, and ESR, as observed in the female Wistar rats studied.
